SHYAM KUMAR REDDY KALLA

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EXPERIENCE

AltiusHub **Nov 2024 – Present**

Software Development Engineer 1 | Python, Django, TypeScript, React, PostgreSQL, Celery, Azure

- Architected microservice-based track-and-trace compliance from scratch for **6** markets (Saudi Arabia, Indonesia, Azerbaijan, Egypt, UAE, Bahrain), automating **26** regulatory report types per country; shipped one country per sprint for **4** enterprise customers.
- Led a **3-engineer** pod (code reviews, mentoring), consistently shipping **3+ features per three-week release**.
- Integrated the platform with Microsoft Dynamics 365 ERP, syncing **5K records/day** and saving about **2 hrs/day** of manual work for **3** enterprise clients.
- Built Smart Migrate, a Django management command that resolves cross-environment migration drift in GitHub Actions deployments (**21** edge cases), cutting DB migration time by **95%**.
- Cut p95 query latency from **3s to 2.4s** on a **100M-row** PostgreSQL table via partitioning, targeted indexes, and ORM query tuning (`select_related`/`prefetch_related`).
- Built serial-number scan pre-validation that blocks invalid traceability events upstream, cutting operator rework by **60%**.

Rockfort Labs **Jul 2023 – Dec 2023**

Machine Learning Intern | Python, Deep Learning

- Built predictive-maintenance anomaly-detection models for industrial boilers on real-time IIoT sensor streams, benchmarking **8** approaches and ranking them by precision/recall tradeoff.
- Productionized evaluation workflows (train/validation splits, thresholding, error analysis) to cut false alarms, reaching **98%** accuracy per boiler on a held-out set.

PROJECTS

Pulse - Distributed Notification System | Python, Django/DRF, Celery, Redis, PostgreSQL, Docker, Kubernetes

- Architected a multi-channel (Email/SMS/Push) notification platform on Django/DRF, Celery, and Redis with priority queues and exponential-backoff retries; sustained **15K+ jobs/min** in Locust load tests.
- Eliminated duplicate sends with idempotency keys, Redis rate limiting (10 req/min per user/channel), and dead-letter queues, validated by **50+** tests across **24** failure and retry scenarios.
- Scaled horizontally with Docker and Kubernetes HPA (2–10 replicas at 50% CPU), holding a **<1% error rate** under **200+** concurrent users in 10-minute stress runs.
- Instrumented observability with Prometheus, Flower, and a Streamlit dashboard for live queue, retry, and failure metrics; scheduled recurring workflows with Celery Beat.

Bhagavad Gita App | TypeScript, React Native, Expo, GitHub Actions

- Built an offline-first, **4-language** (Telugu, English, Hindi, Tamil) scripture app, covering all **18 chapters** and **19 topic categories** with streaks, saved verses, and share cards, with no backend or account.
- Engineered daily-engagement features: a rolling **7-day** local-notification scheduler with tap-to-verse deep links, and a native Android home-screen widget (Kotlin AppWidget with a JS bridge) showing a deterministic verse of the day.
- Enforced quality with Jest catalog-integrity tests, ESLint, and a GitHub Actions workflow that builds release APKs on tag pushes and PRs, backed by documented architecture conventions.

TECHNICAL SKILLS

Languages: Python, Go, TypeScript/JavaScript, Java, SQL
Frameworks/Platforms: Django, Django REST Framework, React, React Native, Expo, Celery, Celery Beat
Data/Infra/Tools: PostgreSQL, Redis, Docker, Kubernetes, Git, Azure, Prometheus, Flower, Streamlit

EDUCATION

SASTRA Deemed University **2020 – 2024**

B.Tech, Computer Science (AI & Data Science) ***CGPA: 8.11*** *Thanjavur, Tamil Nadu*

ACHIEVEMENTS

- Solved **1050+** DSA problems on LeetCode (**Knight**; max rating **1854**; top **5%**).